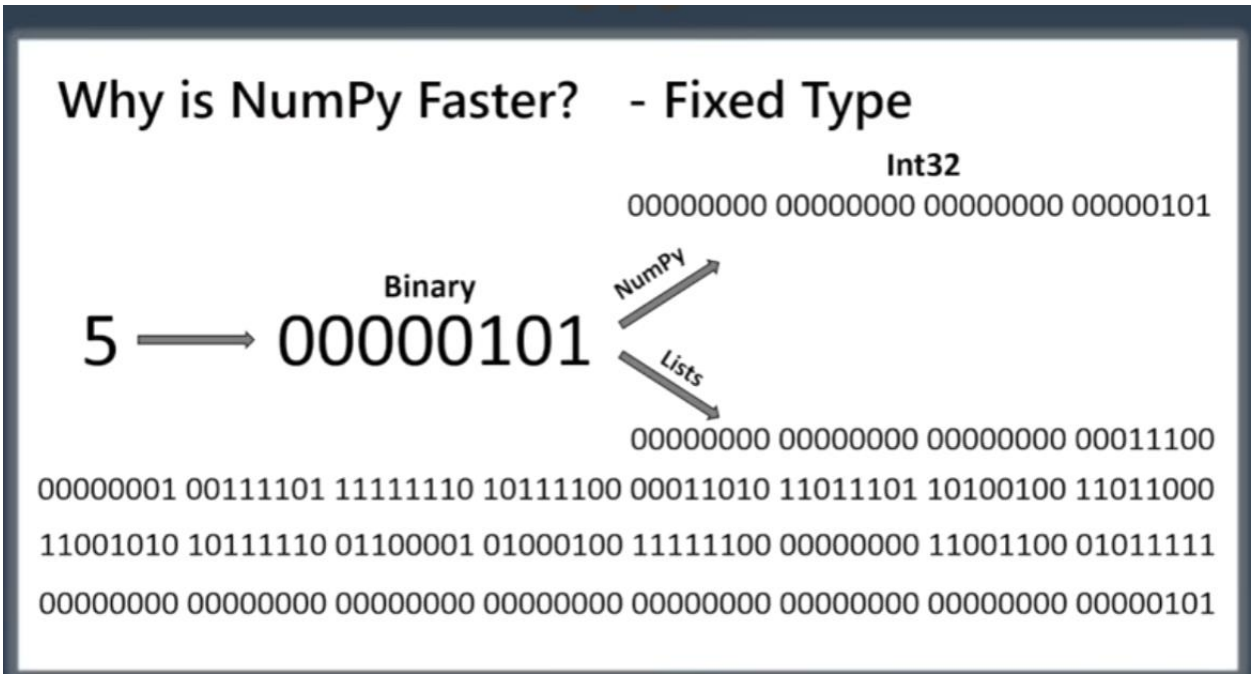
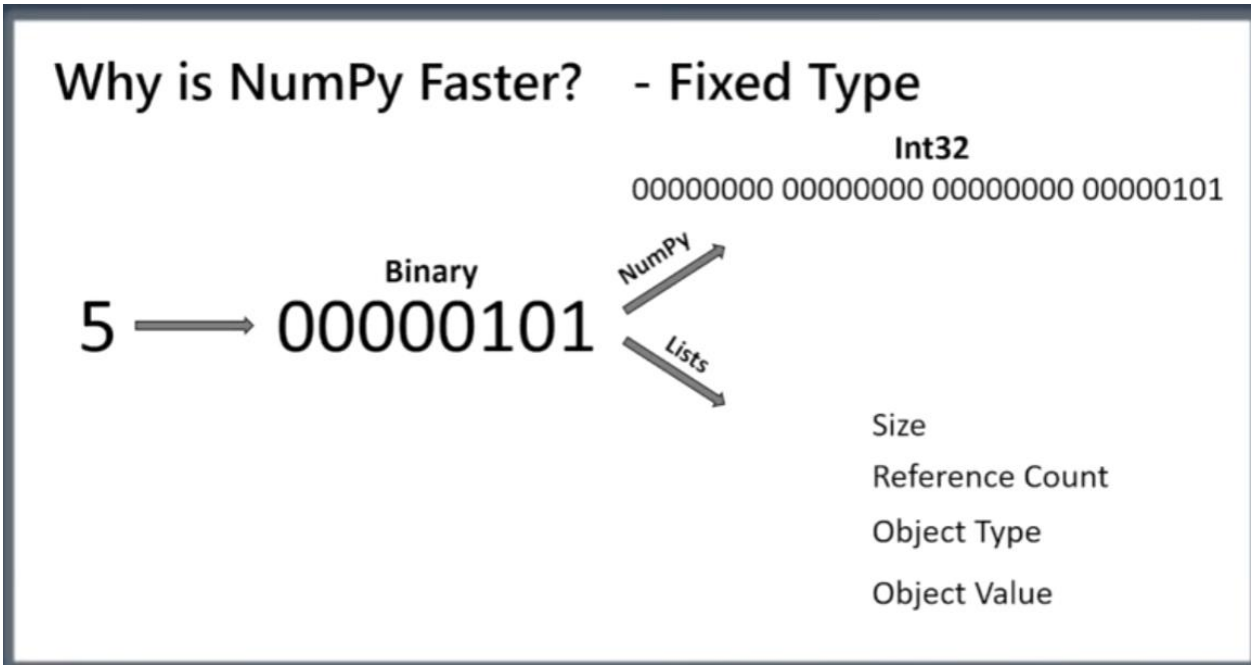
**Question 1:** If the dataset becomes very large (for example, hundreds of thousands or millions of samples), why would a NumPy-based implementation be significantly faster and more efficient than using pure Python loops?

ans: as in numpy ,that specific data is only casted to 8 bits or 32 based on the data specified , while in lists and using loops store multiple things within a specific data { object value : which is the same stored as numpy , reference count : counts how many times these data been pointed at , size of this data , and its type }



So numpy uses less memory , beside when we iterating through each element in a numpy array , we don't check its type whether its integer ,float  etc ….

And it uses contiguous memory which is not scattered like lists .

**Question 2:** If some features are highly correlated or duplicated (multicollinearity), what mathematical problem can occur in the Normal Equation?

There are two basic kinds of multicollinearity:

- **Structural multicollinearity**: This type occurs when we create a model term using other terms. In other words, it's a byproduct of the model that we specify rather than being present in the data itself. For example, if you square term X to model curvature, clearly there is a correlation between X and $X^2$.
- **Data multicollinearity**: This type of multicollinearity is present in the data itself rather than being an artifact of our model. Observational experiments are more likely to exhibit this kind of multicollinearity.

Multicollinearity causes the following two basic types of problems:

- The coefficient estimates can swing wildly based on which other independent variables are in the model. The coefficients become very sensitive to small changes in the model.
- Multicollinearity reduces the precision of the estimated coefficients, which weakens the statistical power of your regression model. You might not be able to trust the p-values to identify independent variables that are statistically significant.

Imagine you fit a regression model and the coefficient values, and even the signs, change dramatically depending on the specific variables that you include in the model. It's a disconcerting feeling when slightly different models lead to very different conclusions. You don't feel like you know the actual effect of each variable!

Why does this happen ?

- The coefficient estimates can swing wildly based on which other independent variables are in the model. The coefficients become very sensitive to small changes in the model.

- Multicollinearity reduces the precision of the estimated coefficients, which weakens the statistical power of your regression model. You might not be able to trust the p-values to identify independent variables that are statistically significant.

References :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